

Forces Investigation — Answer Key

Forces and Motion • Level 4

A. Test results — a toy car was released from the same ramp on different surfaces.

Smooth floor: 120 cm (fast) • Grass: 40 cm (slow) • Carpet: 60 cm (slow)

Answer: Reference data used to answer Section B and C.

B. Answer the questions using the results.

1. Which surface let the toy car travel the farthest?

Answer: The smooth floor (120 cm).

2. Which surface had the most friction?

Answer: Grass — the car travelled the shortest distance (40 cm), showing the most friction.

3. Which surface had the least friction?

Answer: The smooth floor — the car travelled the farthest (120 cm), showing the least friction.

4. Which surface would be best for a racing toy car?

Answer: The smooth floor, because it has the least friction and lets the car travel fastest and farthest.

C. True or False?

1. The car travelled farther on the smooth floor.

Answer: True

2. Grass has less friction than a smooth floor.

Answer: False — grass has more friction than a smooth floor.

3. Friction can affect how far something moves.

Answer: True

4. A fair test changes one thing at a time.

Answer: True

D. Why do scientists test surfaces and forces?

Answer: To compare results fairly, find patterns (like which surfaces have more or less friction), and use what they learn to predict and explain how things will move in other situations.